LLMOps Assignment 2: API Cost, Latency and Prompt Version Management

Course: MLOps | EduAssist AI scenario

1. Introduction

EduAssist AI needs an academic question-answering assistant that returns strict JSON. This report measures three prompt versions using gemini-3.5-flash-lite (Gemini API v1beta (OpenAI-compatible endpoint)), priced at 0.3 USD per million input tokens and 2.5 USD per million output tokens, as published on 2026-11-03. The 5 questions come from MTech modules and mimic a real support desk (definition, debugging, mlops scenario, architecture, llmops economics), so answer lengths differ. Each version was sent every question 3 times, interleaved, giving 45 requests at temperature 0.0; all succeeded.

2. Task 1: Latency

For the five v1 requests of the first repeat, mean latency was 2.94s, median 3.02s, minimum 2.19s, maximum 3.82s and standard deviation 0.69s. Across all 45 requests the slowest was Q1 with prompt v1 at 4.43s and 519 output tokens. Latency closely tracks output length (r = 0.86): generation is sequential, but network and provider load add noise. Repeating requests quantified it: mean latency with p95 was v1 2.95s (p95 4.01s), v2 2.04s (p95 2.59s), v3 1.31s (p95 2.15s). Relative to v3, the other versions were slower by 1.64s (6.4 standard errors) for v1, 0.73s (4.0 standard errors) for v2; gaps under about two standard errors are not reliable, so latency alone should not decide the promotion.

question_id	prompt_version	start_time	end_time	latency_seconds	status
Q1	v1	2026-11-03T20:37:14.970+00:00	2026-11-03T20:37:17.985+00:00	3.015	success
Q2	v1	2026-11-03T20:37:17.986+00:00	2026-11-03T20:37:20.175+00:00	2.19	success
Q3	v1	2026-11-03T20:37:20.177+00:00	2026-11-03T20:37:24.001+00:00	3.825	success
Q4	v1	2026-11-03T20:37:24.003+00:00	2026-11-03T20:37:26.325+00:00	2.322	success
Q5	v1	2026-11-03T20:37:26.326+00:00	2026-11-03T20:37:29.682+00:00	3.357	success

3. Task 2: Tokens and cost

Cost per request is input tokens / 1,000,000 x input price plus output tokens / 1,000,000 x output price. Average cost per request was v1: 0.001637, v2: 0.001020, v3: 0.000448 USD, which projects to v1: 16.37, v2: 10.20, v3: 4.48 USD for 10,000 requests a month. Output tokens cost 8.3 times as much as input tokens and make up 93% to 99% of spend, so the most effective saving is to shorten responses: v3 wrote 74% fewer output tokens than v1 and cost 73% less. Capping max tokens, caching the stable prompt prefix and routing simple questions to a smaller model are further options.

question_id	prompt_version	input_tokens	output_tokens	total_tokens	input_cost	output_cost	total_cost
Q1	v1	28	671	699	8e-06	0.001678	0.001686
Q2	v1	43	450	493	1.3e-05	0.001125	0.001138
Q3	v1	50	767	817	1.5e-05	0.001918	0.001932
Q4	v1	25	507	532	8e-06	0.001268	0.001275
Q5	v1	37	831	868	1.1e-05	0.002078	0.002089

4. Business and scale view

The brief fixes 10,000 requests a month, but a real service sees other volumes. At the campus-wide volume of 1,000,000 requests a month, v3 costs 448 USD against 1,637 for v1, saving 1,190 a month.

Against an assumed 2s interactive target, the share of requests within it was v1: 0%, v2: 53%, v3: 93%. At an assumed peak of 5 requests per second, the service needs about v1: 14.7, v2: 10.2, v3: 6.6 requests in flight. Volumes, target and peak rate are illustrative assumptions, not measurements.

version	Baseline (brief) (USD/month)	Exam season (USD/month)	Campus-wide (USD/month)	concurrent_requests_at_peak
v1	16.4	163.7	1637.3	14.7
v2	10.2	102.0	1020.2	10.2
v3	4.5	44.8	447.6	6.6

5. Task 3: Prompt versioning and promotion

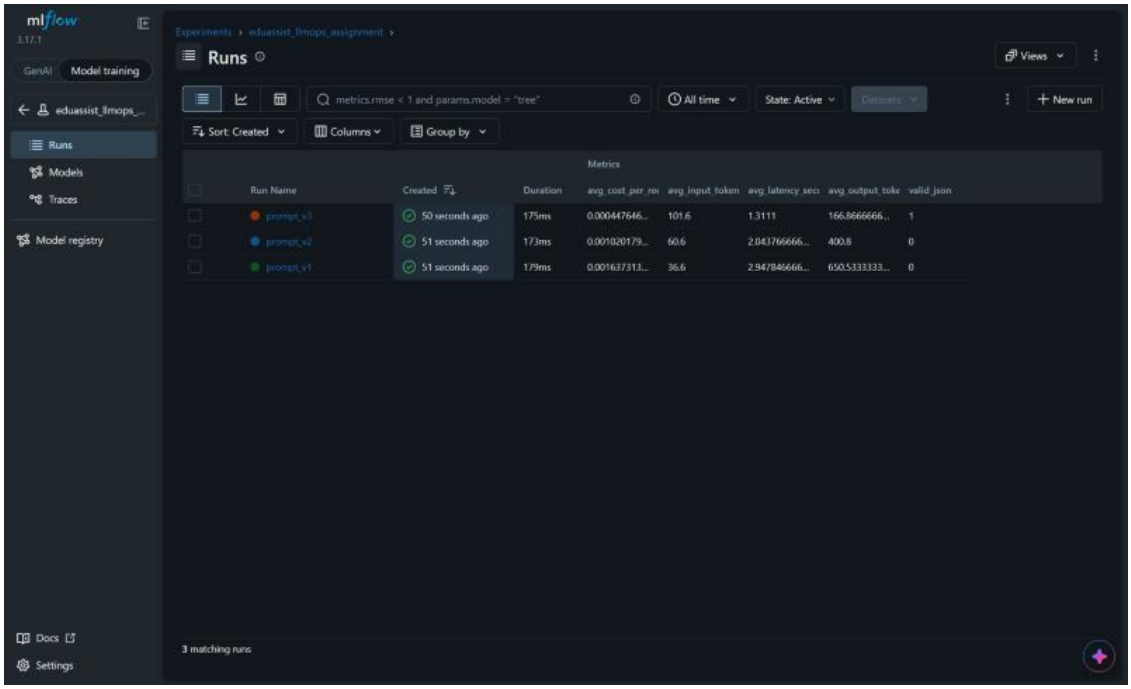
The prompts were registered as versions of eduassist-qa-prompt in the MLflow Prompt Registry with commit messages, evaluated on the same questions, and logged in eduassist_llmops_assignment. Version v3 was promoted by setting the production alias. Promotion justification: the product contract requires strict JSON, so validity decides. Valid JSON rates were v1: 0%, v2: 0%, v3: 100%. Version v3 also had the lowest average latency (1.31s), the fewest output tokens (167) and the lowest cost (0.000448 USD per request). It sends more input tokens because the schema is included, but input is the cheap side. A free-text answer would need a second model call or fragile parsing to become JSON. Answer correctness was not scored automatically; the confidence and key_points fields support human review. The alias lets the application load the production prompt without code changes and roll back instantly.

prompt_version	avg_latency_seconds	avg_input_tokens	avg_output_tokens	avg_cost_per_request	valid_json	recommended
v1	2.947847	36.6	650.533333	0.001637	0.0	No
v2	2.043767	60.6	400.8	0.00102	0.0	No
v3	1.3111	101.6	166.866667	0.000448	1.0	Yes

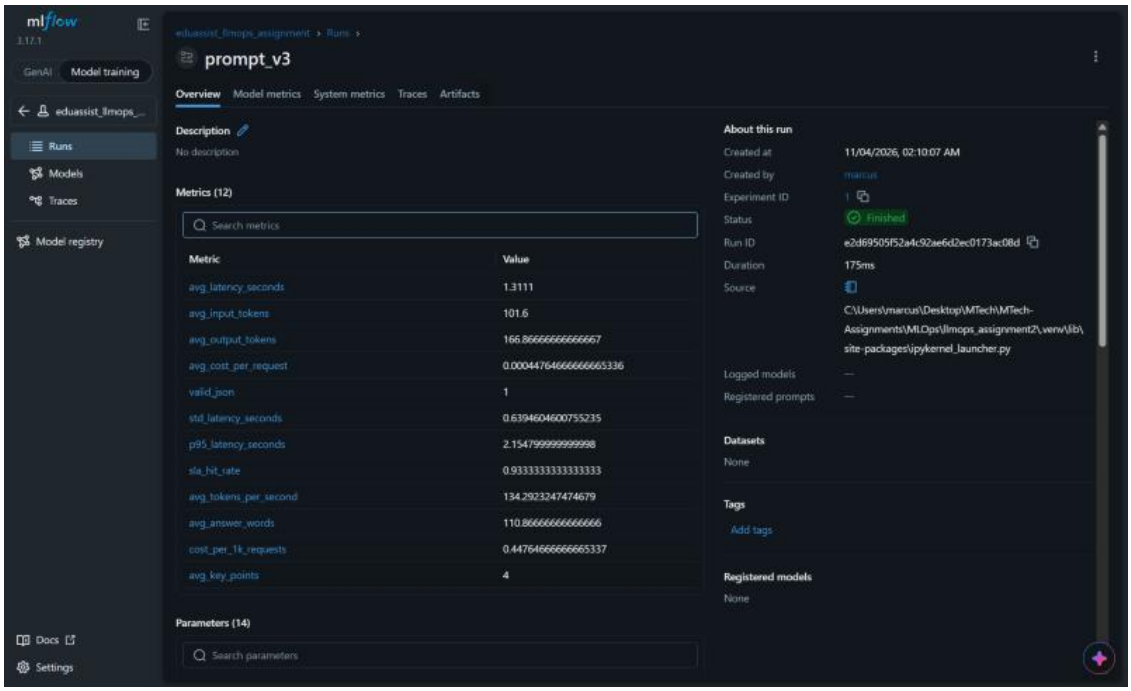
6. Conclusion

Treating prompts as versioned artifacts, with measured latency, tokens and cost, gives a reproducible basis for promotion. Limitations: five questions is a small sample, the scale figures rest on stated assumptions, and quality scoring on a larger set should confirm the choice before wide rollout.

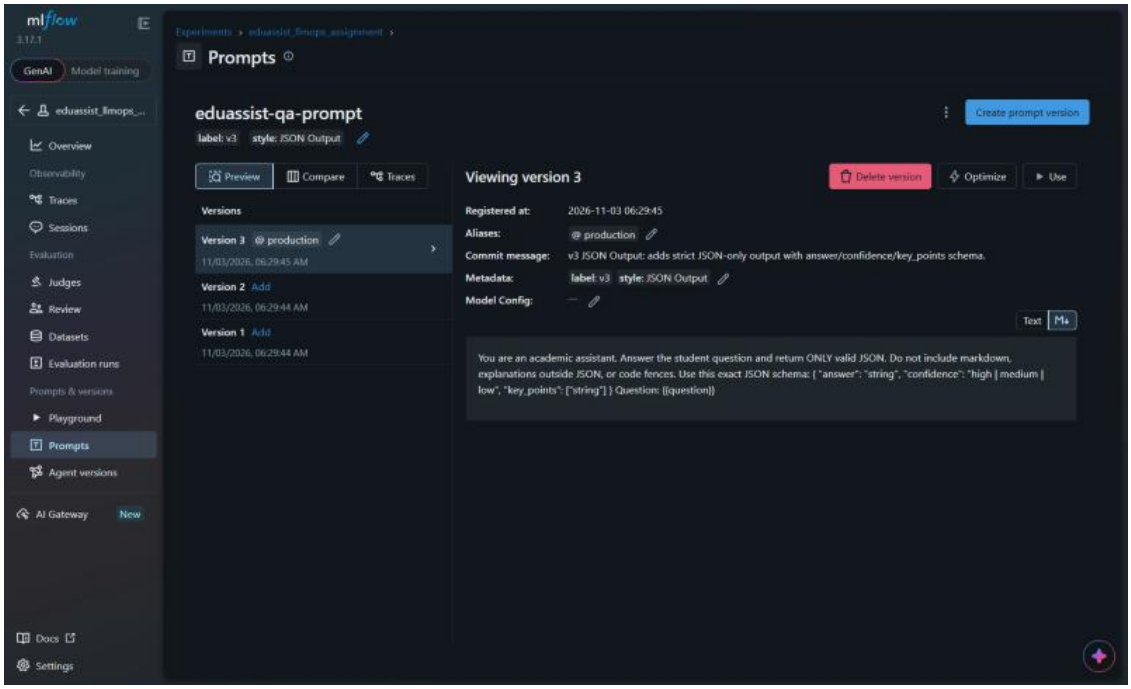
MLflow experiment: runs and metrics



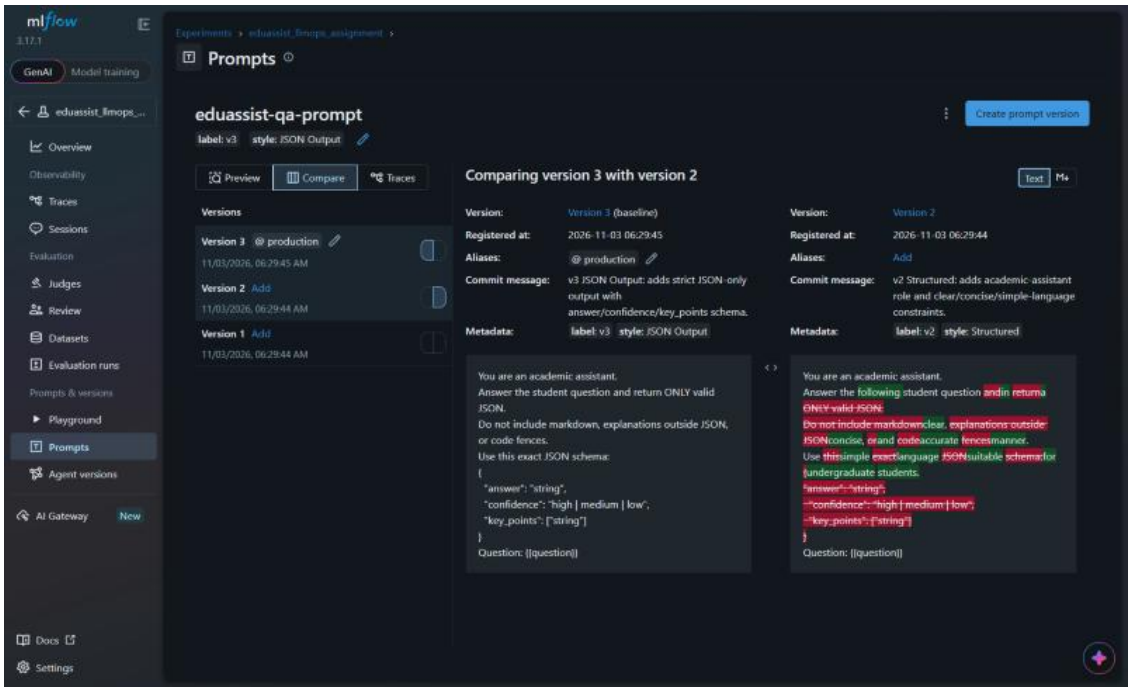
Run metrics (prompt_v3)



Prompt Registry: v1, v2, v3



Prompt diff (v2 vs v3)



`production` alias on the promoted version

